Do Not Read This

Read This

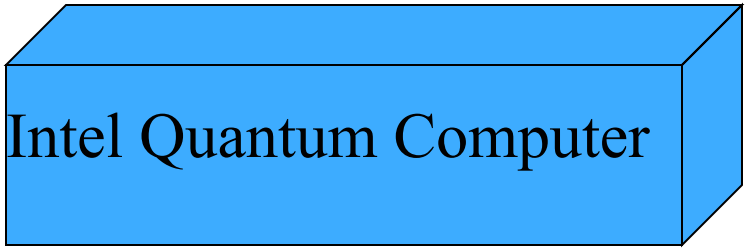
sihT esraP oT yrT

Introduction to Quantum Error Correction

or

Why We're Still Here, Dr. Landauer





Dave Bacon

Institute For Quantum Information

Caltech

A Physicist Talking About Error Correction?

Phthisis (ˈfθaɪsɪs, ˈtaɪsɪs). A progressive wasting disease; spec. pulmonary consumption.

A Critical Ghost



Rolf Landauer
IBM

All papers on quantum computing should carry a footnote: *"This proposal, like all proposals for quantum computation, relies on speculative technology, does not in its current form take into account all possible sources of noise, unreliability and manufacturing error, and probably will not work."*

- Maintenance of giganto-coherence?
- Faulty quantum gates?
- Do we understand the physics of quantum errors in the system?

Nature abhors a quantum computer?





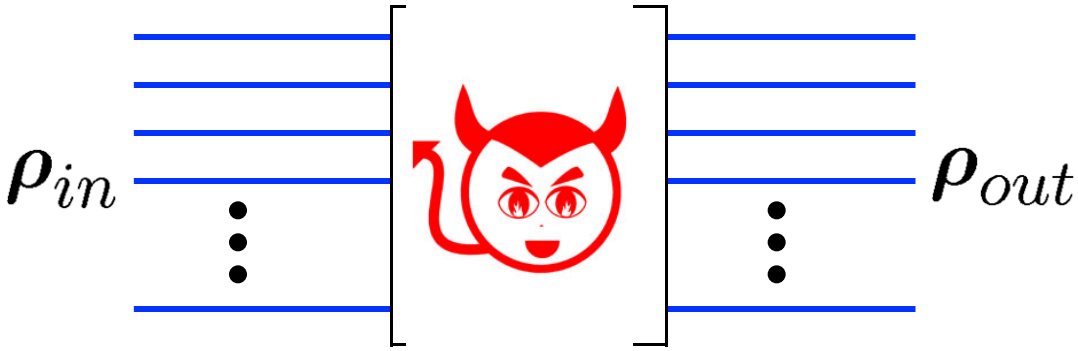
Quantum
Error
Correction!



Rolf Landauer
IBM

Threshold Theorem for Fault-Tolerant Quantum Computation

Don't Eat That Apple



plus: simple
minus: unrealistic
plus: essential ideas

Lucifer's channel:

set of unitaries: $\mathcal{S} = \{\mathbf{U}_1 = \mathbf{I}, \mathbf{U}_2, \mathbf{U}_3, \ldots, \mathbf{U}_k\}$

probabilities: $p_1, p_2, p_3, \ldots, p_k$, $p_i \geq 0$, $\sum_i p_i = 1$

devil applies unitary $\mathbf{U}_i$ with probability p_i

no which $\mathbf{U}_i$ information available

$$\boldsymbol{\rho}_{out} = \sum_i p_i \mathbf{U}_i \boldsymbol{\rho}_{in} \mathbf{U}_i^\dagger$$

Identity

simplest example (single qubit):

$$\mathcal{S} = \{\mathbf{I}, \mathbf{X}, \mathbf{Z}\},\ p_i > 0$$

$$\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

How to preserve $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$?

less silly? example (two qubits):

$$\mathcal{S} = \{\mathbf{I}, \mathbf{Z}_1\mathbf{Z}_2\},\ p_i > 0$$

$$\mathbf{Z}_1\mathbf{Z}_2 = \begin{array}{c} \begin{matrix} |00\rangle & |01\rangle & |10\rangle & |11\rangle \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

notation

$$\mathbf{P}_i = \mathbf{I} \otimes \cdots \otimes \mathbf{I} \otimes \underbrace{\mathbf{P}}_{ith\ qubit} \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Encode information into first qubit? $(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle$

$$\mathbf{Z}_1\mathbf{Z}_2(\alpha|0\rangle + \beta|1\rangle)|0\rangle = (\alpha|0\rangle - \beta|1\rangle)|0\rangle$$

Solution: encode into subspace $|00\rangle$, $|11\rangle$

$$\mathbf{Z}_1\mathbf{Z}_2(\alpha|00\rangle + \beta|11\rangle) = \alpha|00\rangle + \beta|11\rangle$$

Qubit can be protected by encoding into a subspace

The Story of the Ghost



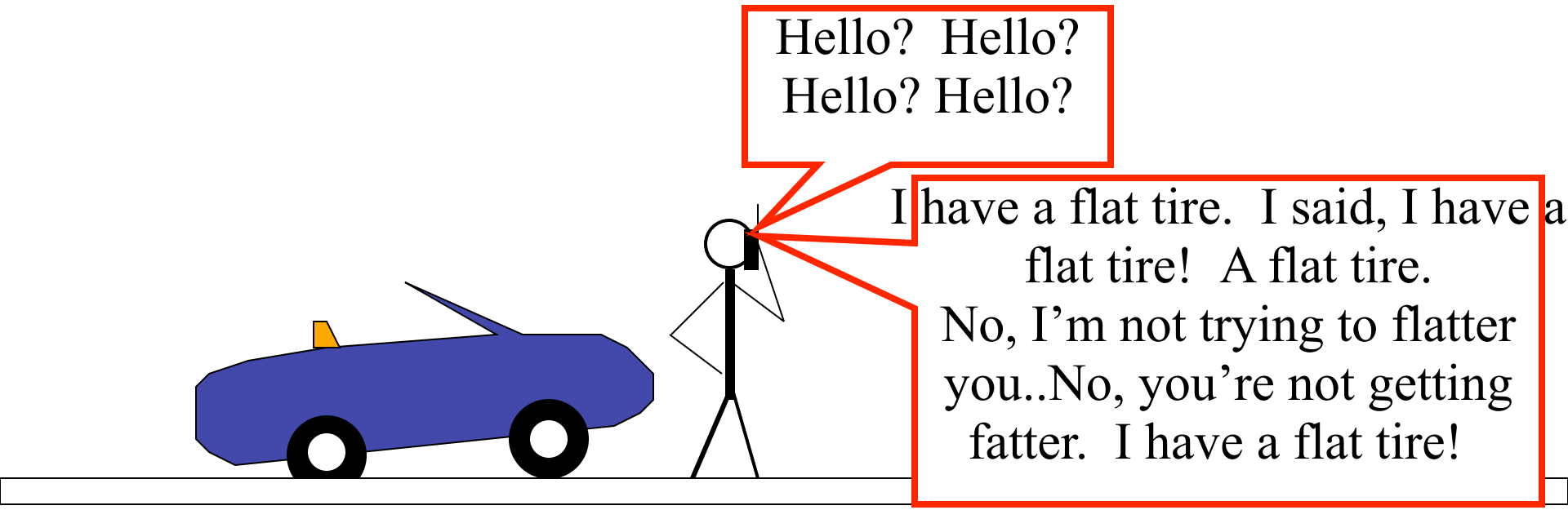


You are protecting your quantum information against a crazy noise model! Z1Z2? If this is all nature can throw at you, then pigs can fly.

Rolf Landauer
IBM

Can we use encoding to protect quantum information from more realistic noise models?

Noisy Cell Phone



Communication over a noisy CHANNEL can be overcome via

ENCODING

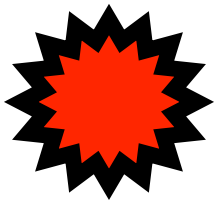
"Hello?" = "Hello? Hello? Hello? Hello?" [using redundancy to encode "Hello"]

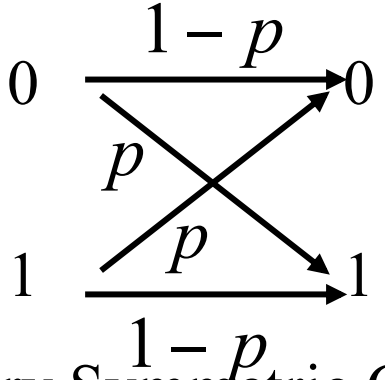
Simple Repetition Code

Encode: $0 \rightarrow 000\ldots000$

$1 \rightarrow 111\ldots111$

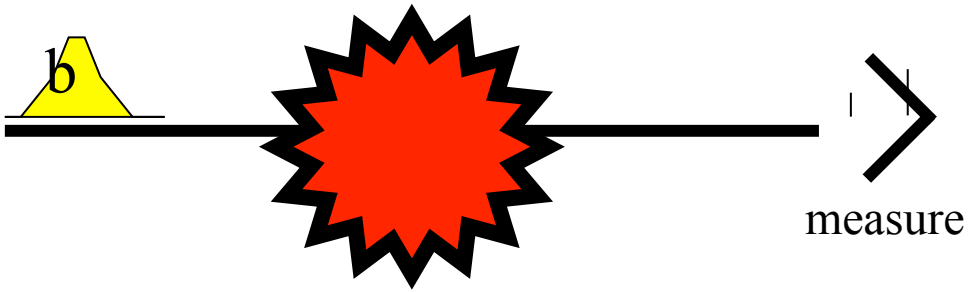
n copies





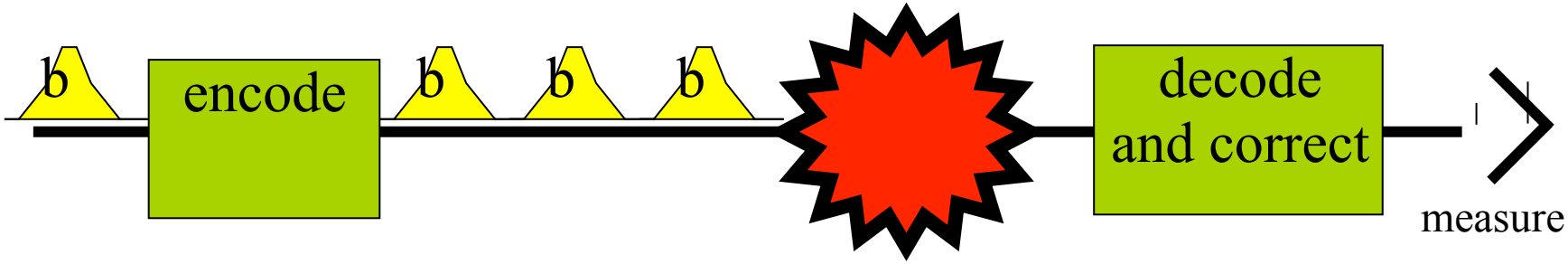
Binary Symmetric Channel

No encoding:



Probability of error = p

Encoding (n=3):



Probability of error $= 3p^2(1-p) + p^3 = 3p^2 - 2p^3 < p, p < \frac{1}{2}$

1994 Reasons to be a Pessimist

No cloning:



Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

No quantum repetition code:

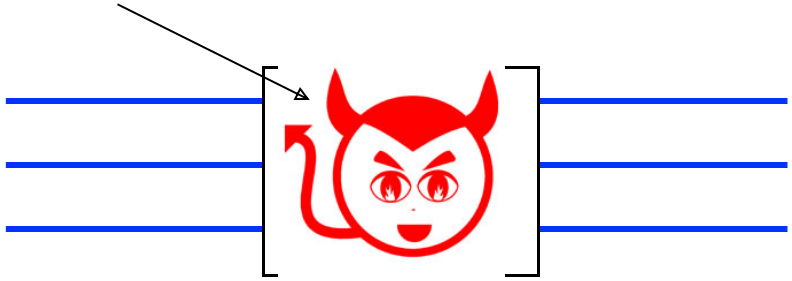
$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

Measurement destroys coherence:

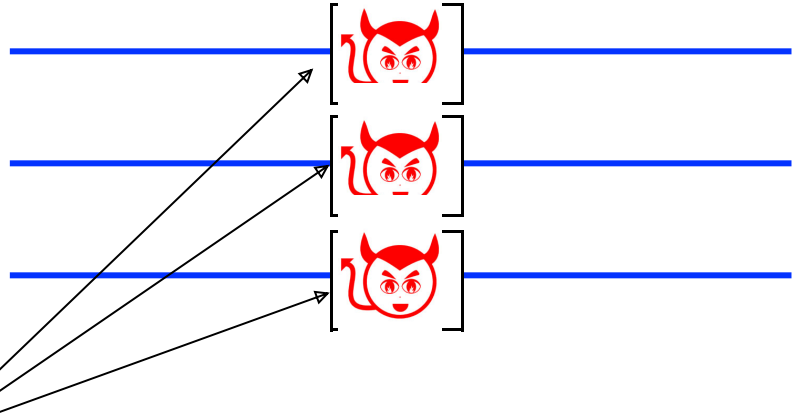
How can one decode without destroying the information?

Unrealistic Realistic Channel

$\mathcal{S} = \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3\},\ p_1 = 1 - 3\epsilon, p_2 = p_3 = p_4 = \epsilon$



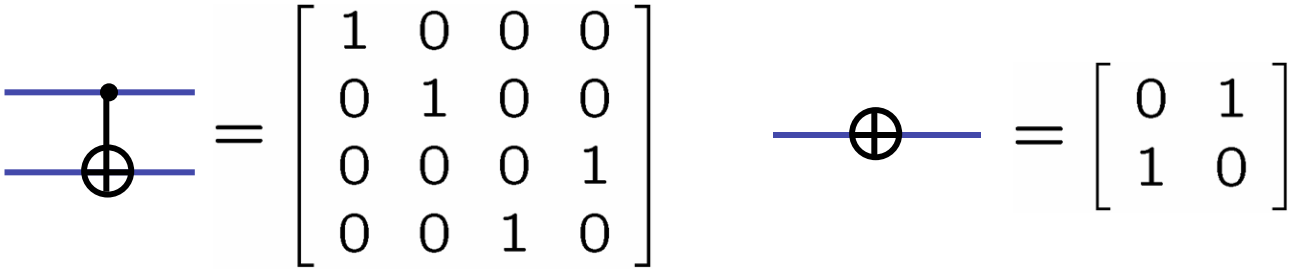
≀



$\mathcal{S} = \{\mathbf{I}, \mathbf{X}\},\ p_1 = 1 - \epsilon, p_2 = \epsilon$

Baby Steps



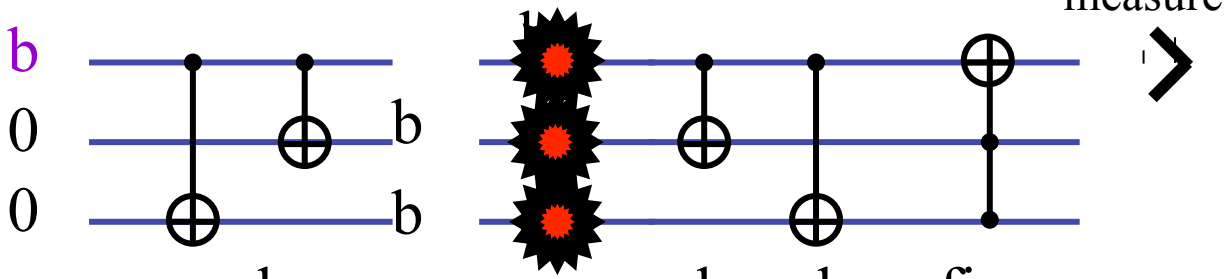


$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

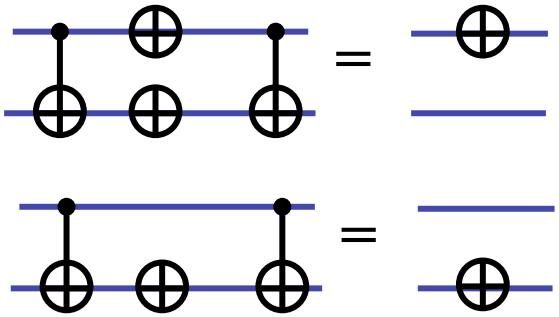
WWCCD? (What Would Classical Coders Do?)

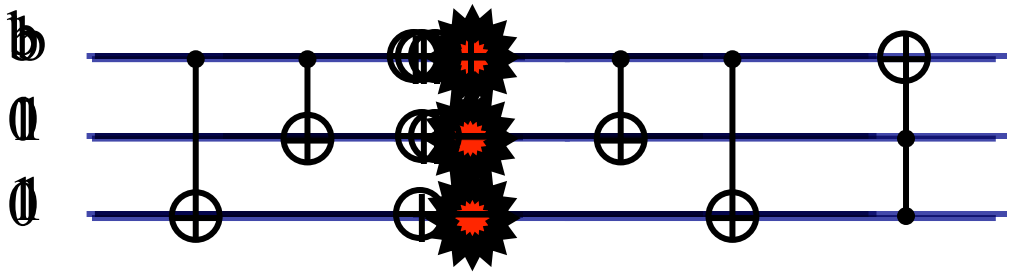
$0 \rightarrow 000,\ 1 \rightarrow 111$



100 → 111→101 → 110 → **1**10

identities



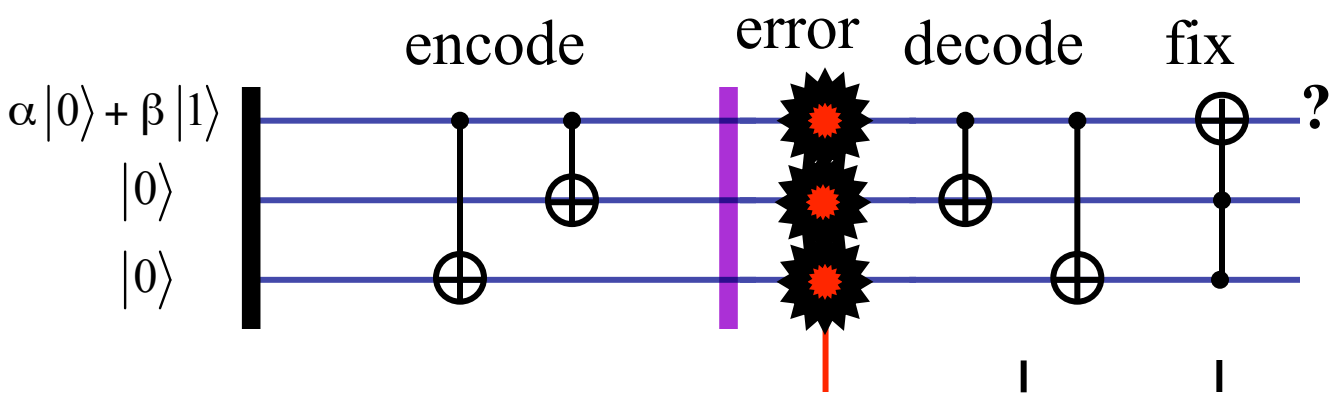


error #@%

Naïve

4. operator identities still hold

Lets be naïve, take classical and move over to quantum



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|0\rangle \otimes |0\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle \otimes |1\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{XII}{\rightarrow} \alpha|100\rangle + \beta|011\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IXI}{\rightarrow} \alpha|010\rangle + \beta|101\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IIX}{\rightarrow} \alpha|001\rangle + \beta|110\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$

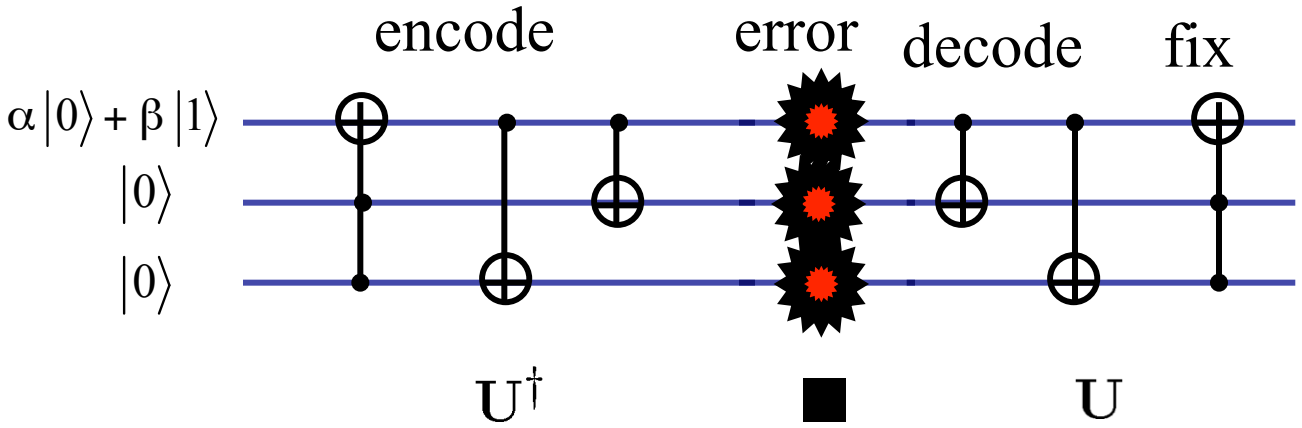
(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

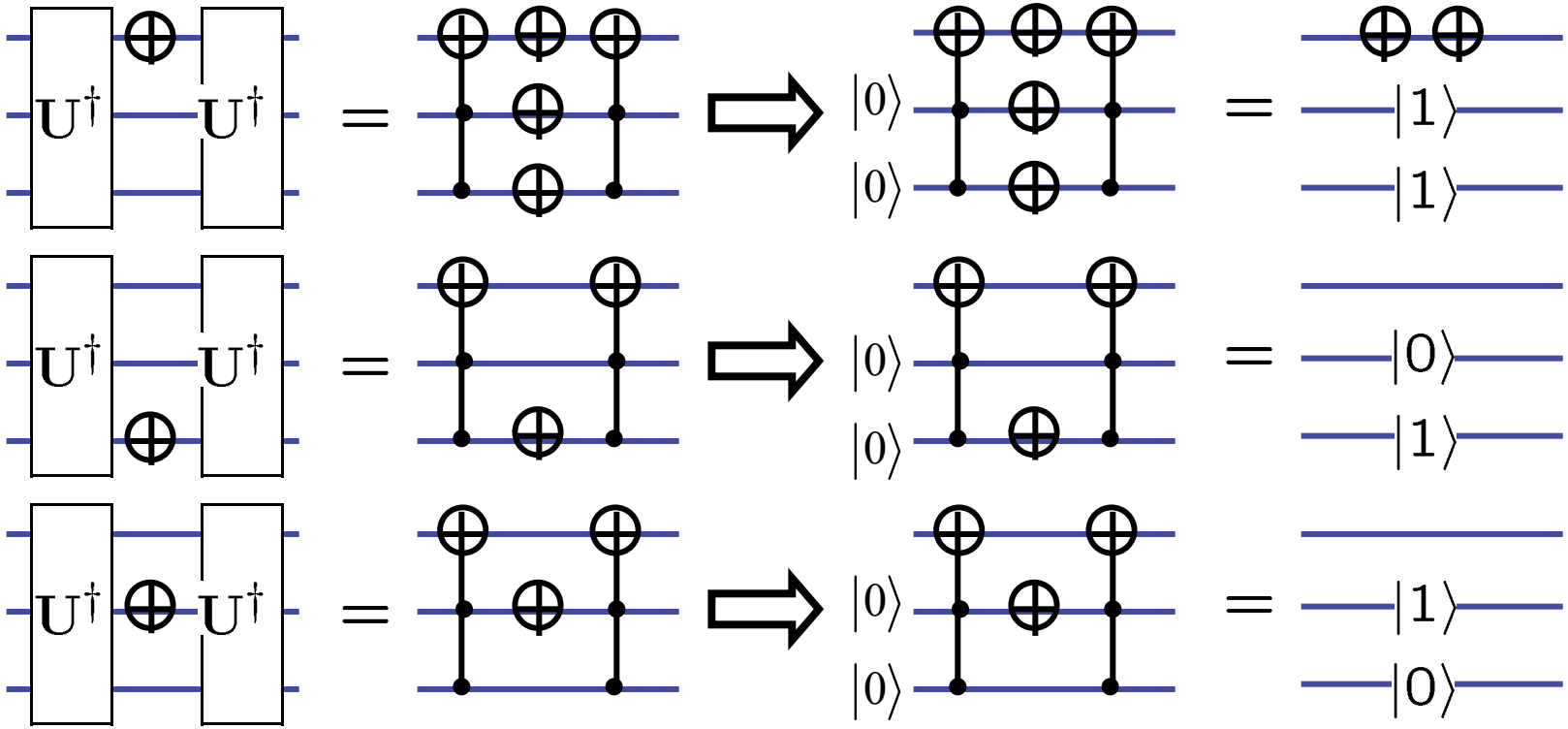
$\mathbf{XII}|000\rangle = |100\rangle, \mathbf{XII}|111\rangle = |011\rangle$

3. syndrome

Identity



But this is just a basis change: $UEU^\dagger$, E is the error



OK Wise Guy

What about "phase" errors? $\rho_{out} = (1-p)\rho_{in} + p\sigma_z\rho_{in}\sigma_z$

phase error: $\sigma_z(\alpha|0\rangle + \beta|1\rangle) = (\alpha|0\rangle - \beta|1\rangle)$ …sort of not classical error

$\mathcal{S} = \{\mathrm{I}, \mathrm{X}_1, \mathrm{X}_2, \mathrm{X}_3\},\ p_1 = 1-3\epsilon, p_2 = p_3 = p_4 = \epsilon$

Wise guy says "basis change please":

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

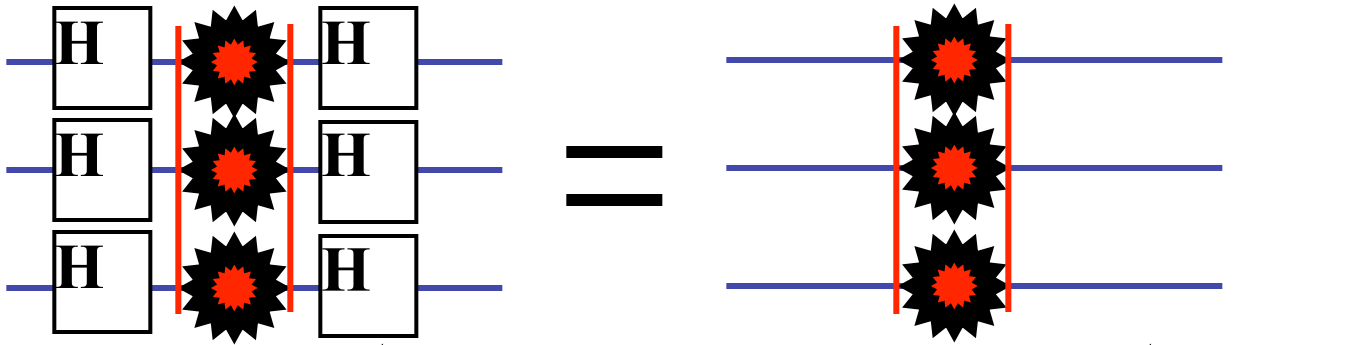
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$\sigma_z|+\rangle = |-\rangle$

$\sigma_z|-\rangle = |+\rangle$

looks like bit flit error in this new basis!

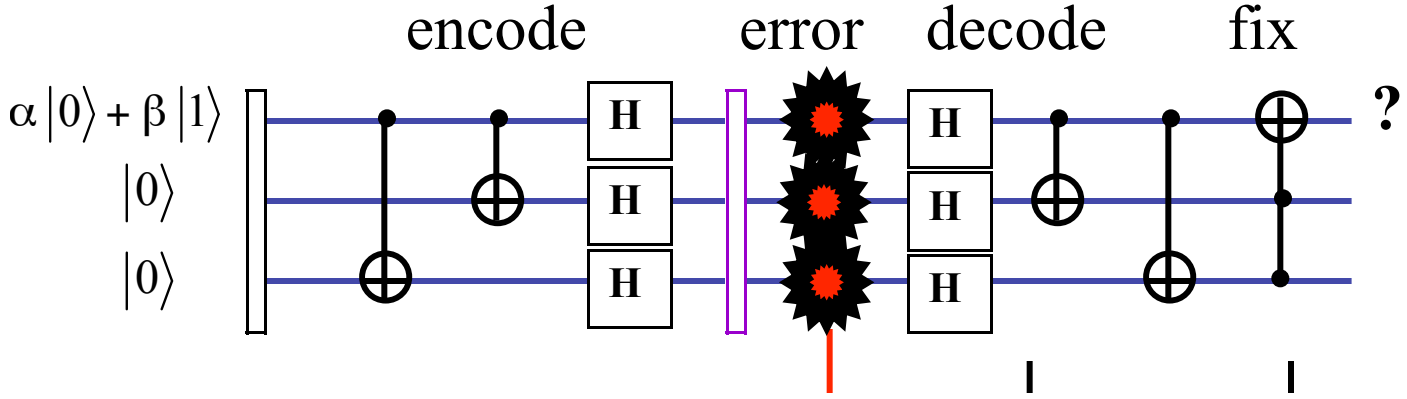
$$\mathrm{H}\sigma_z\mathrm{H} = \sigma_x, \quad \mathrm{H} = |+\rangle\langle 0| + |-\rangle\langle 1|$$



phase errors $= \sigma_z$ bit flip errors $= \sigma_x$

Molly: "I love you, I really love you"

Sam: "Ditto."



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|+\rangle \otimes |+\rangle \otimes |+\rangle + \beta|-\rangle \otimes |-\rangle \otimes |-\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{ZII}{\rightarrow} \alpha|-++\rangle + \beta|+--\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)\,|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|11\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IZI}{\rightarrow} \alpha|+-+\rangle + \beta|-+-\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IIZ}{\rightarrow} \alpha|++-\rangle + \beta|--+\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |+++\rangle, |1\rangle \rightarrow |---\rangle$

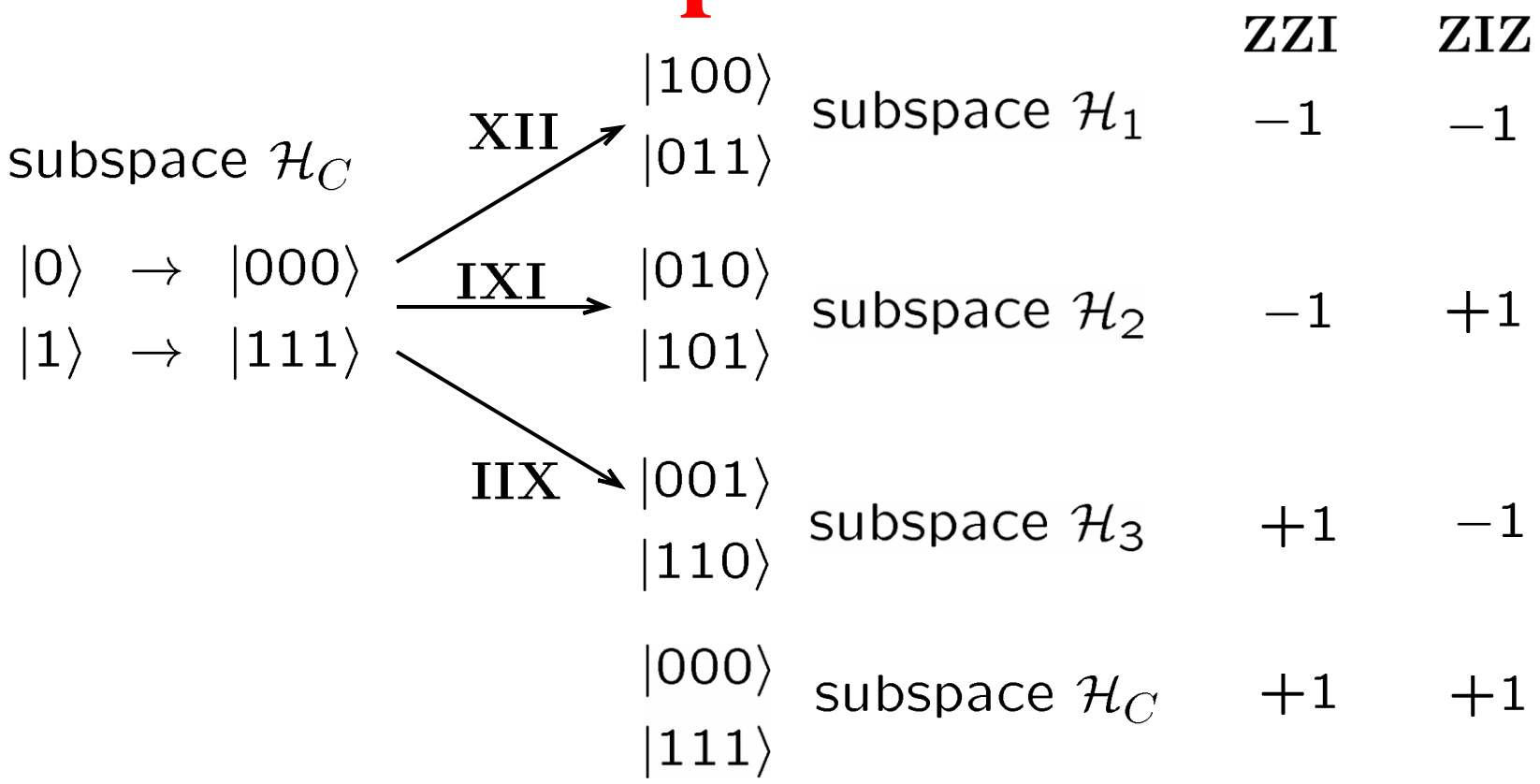
(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

$\mathbf{ZII}|+++\rangle = |-++\rangle, \mathbf{ZII}|---\rangle = |+--\rangle$

3. syndrome

Perspective



subspaces $\mathcal{H}_C$, $\mathcal{H}_1$, $\mathcal{H}_2$, and $\mathcal{H}_3$ are orthogonal

$\dim \mathcal{H}_C = \dim \mathcal{H}_1 = \dim \mathcal{H}_2 = \dim \mathcal{H}_3 = 2$

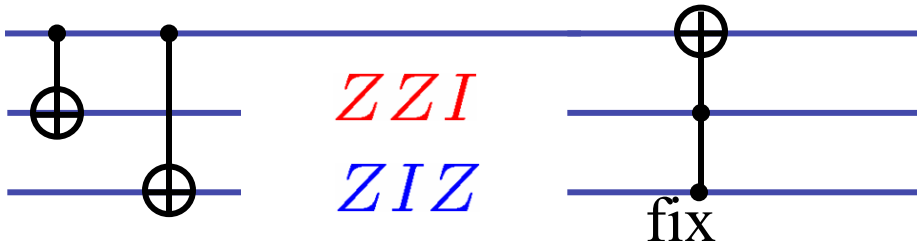
Orthogonal subspaces can be <u>distinguished</u> by measurements

<u>without</u> measuring information encoded into the subspace!!!

Not An Optical Illusion



		b	ZZ
$\|00\rangle$	$\rightarrow$	$\|00\rangle$	$+1$
$\|01\rangle$	$\rightarrow$	$\|01\rangle$	-1
$\|10\rangle$	$\rightarrow$	$\|11\rangle$	-1
$\|11\rangle$	$\rightarrow$	$\|10\rangle$	$+1$



		XII	$\|100\rangle$	$\|111\rangle$	$\|011\rangle$
			$\|011\rangle$	$\|011\rangle$	$\|111\rangle$
$\|0\rangle$	$\rightarrow$ $\|000\rangle$	IXI	$\|010\rangle$	$\|010\rangle$	$\|010\rangle$
$\|1\rangle$	$\rightarrow$ $\|111\rangle$		$\|101\rangle$	$\|110\rangle$	$\|110\rangle$
		IIX	$\|001\rangle$	$\|001\rangle$	$\|001\rangle$
		error	$\|110\rangle$	$\|101\rangle$	$\|101\rangle$

Encoding Away Your Ills

3 qubit bit flip code	3 qubit phase flip code
$\|0\rangle \rightarrow \|000\rangle$	$\|0\rangle \rightarrow \|+++\rangle$
$\|1\rangle \rightarrow \|111\rangle$	$\|1\rangle \rightarrow \|---\rangle$

phase errors **ZII, IZI, IIZ** act as ■ on bit flip code qubits:

$$|0\rangle_B = |000\rangle \qquad \mathbf{ZII}|0\rangle_B = |0\rangle_B$$
$$|1\rangle_B = |111\rangle \qquad \mathbf{ZII}|1\rangle_B = -|1\rangle_B$$

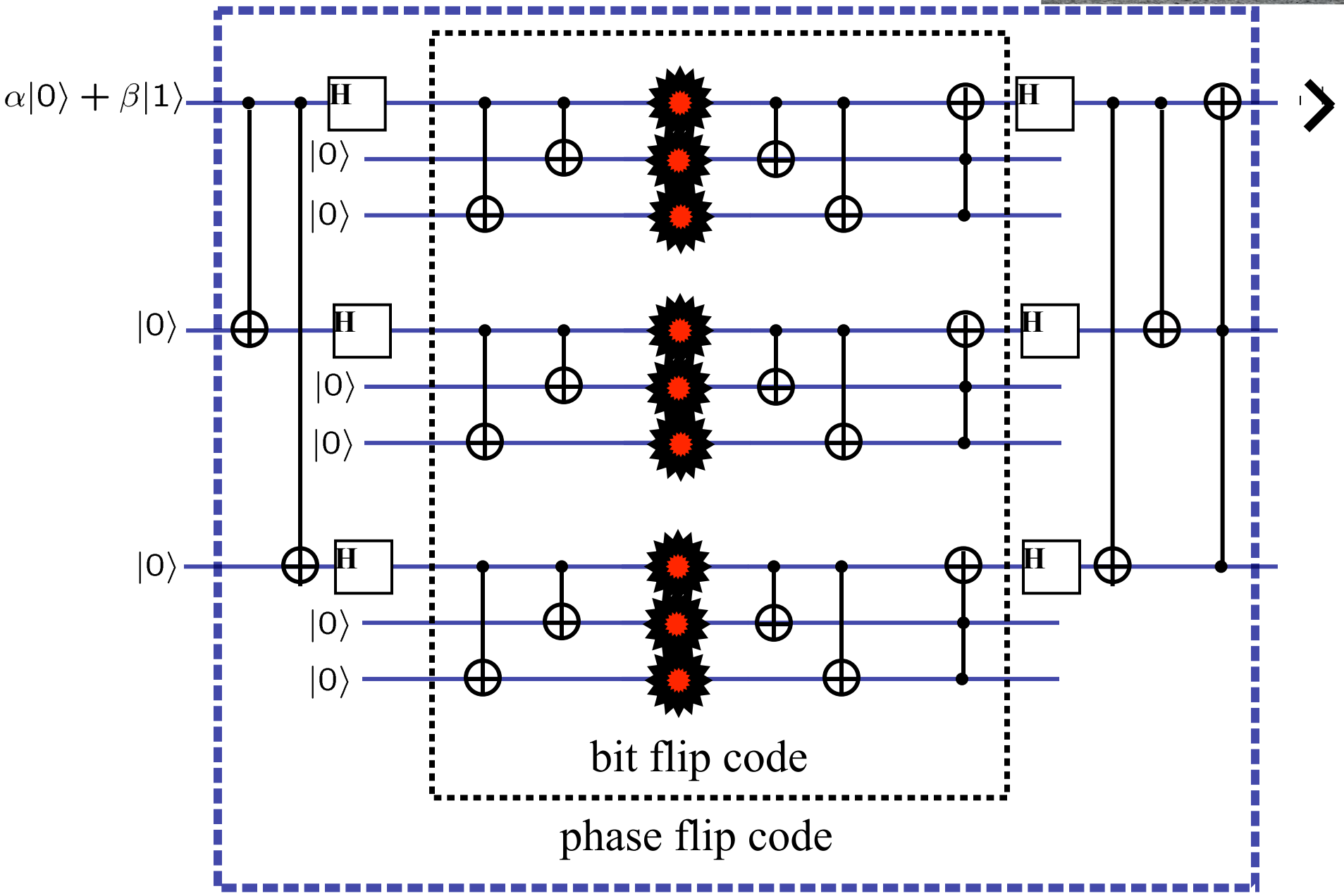
define:
$$|p\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B - |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

Shor Code: (Peter Shor, 1995)

$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$
$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

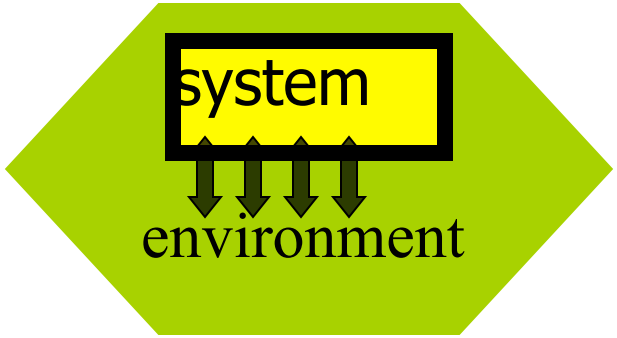
Inside Shor




$\alpha|0\rangle + \beta|1\rangle$
$|0\rangle$
$|0\rangle$
H
H
H
H
H
H
bit flip code
phase flip code

What is the Problem?

Real quantum systems are open quantum systems!



Full Hilbert space $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$

Evolution is $\mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$

Full Hamiltonian:

System evolution — environment evolution — **Interaction causing DECOHERENCE**

$$\mathbf{H}_{SB} = \mathbf{H}_S \otimes \mathbf{I}_B + \mathbf{I}_S \otimes \mathbf{H}_B + \mathbf{H}_I$$

System's pure states become entangled with environment:

$|\psi_0\rangle = |\psi\rangle_S \otimes |\psi\rangle_B \rightarrow \sum_n c_n |n\rangle \otimes |n\rangle \quad \rho_0 = |\psi_0\rangle_S \, {}_S\langle\psi_0| \rightarrow$ mixed state

System evolution (by itself) is no longer unitary

Coherent properties of system's quantum information lost

Quantum Error Example

"To err is human, to really foul things up requires a computer." -Anonymous

$$\mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$$



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \rightarrow \alpha|00\rangle + \beta|11\rangle$$

$$\boldsymbol{\rho}_{sys} = \mathrm{Tr}_{env}\left[\boldsymbol{\rho}_{full}\right] = \langle 0_{env}|\boldsymbol{\rho}_{full}|0_{env}\rangle + \langle 1_{env}|\boldsymbol{\rho}_{full}|1_{env}\rangle$$

Before $\mathbf{U}$: $\boldsymbol{\rho}_{sys} = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$ $\quad \mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$

After $\mathbf{U}$: $\boldsymbol{\rho}_{sys} = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$

Not unitary:

$$\det\left[\mathbf{U}\boldsymbol{\rho}\mathbf{U}^\dagger\right] = \det\left[\mathbf{U}\right]\det\left[\mathbf{U}^\dagger\right]\det\left[\boldsymbol{\rho}\right] = \det\left[\boldsymbol{\rho}\right]$$

Operator Sum Representation

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$



Assume initially $\rho = \rho_A \otimes \rho_B$

$$\rho_{out} = \mathrm{Tr}_B [\rho_{AB}] = \mathrm{Tr}_B \left[\mathbf{U} (\rho_A \otimes \rho_B) \mathbf{U}^\dagger \right]$$

Spectrally decompose ρ_B: $\rho_B = \sum_\nu p_\nu |\nu\rangle\langle\nu|$

$$\begin{aligned} \rho_{out} &= \sum_\nu p_\nu \mathrm{Tr}_B \left[\mathbf{U} (\rho_A \otimes |\nu\rangle\langle\nu|) \mathbf{U}^\dagger \right] \\ &= \sum_{\nu,\mu} p_\nu \langle\mu| \mathbf{U} (\rho_A \otimes |\nu\rangle\langle\nu|) \mathbf{U}^\dagger |\mu\rangle \\ &= \sum_{\nu\mu} \sqrt{p_\nu} \langle\mu|\mathbf{U}|\nu\rangle \rho_A \langle\nu|\mathbf{U}^\dagger|\mu\rangle \sqrt{p_\nu} \end{aligned}$$

B states

$\rho_{out} = \sum_k \mathbf{A}_k \rho_A \mathbf{A}_k^\dagger$ where $\mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu} \langle\mu|\mathbf{U}|\nu\rangle$

Exercise: Unitarity of $\mathbf{U}$ implies $\sum_k \mathbf{A}_k^\dagger \mathbf{A}_k = \mathbf{I}$

OSR Facts

$$\boldsymbol{\rho}' = \$[\boldsymbol{\rho}] = \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger \qquad\qquad \sum_k \mathbf{A}_k^\dagger \mathbf{A}_k = \mathbf{I}$$

Properties of these maps $\$: \boldsymbol{\rho} \to \boldsymbol{\rho}'$

- $\$ preserves hermiticity: if $\boldsymbol{\rho}$ is hermitian, $\$[\boldsymbol{\rho}]$ is hermitian.
- $\$ is trace preserving: $\mathrm{Tr}\,[\$[\boldsymbol{\rho}]] = \mathrm{Tr}\,[\boldsymbol{\rho}]$
- $\$ is positive: $\$[\boldsymbol{\rho}]$ is positive if $\boldsymbol{\rho}$ is positive.
- $\$ is linear: $\$[\alpha\mathbf{A} + \beta\mathbf{B}] = \alpha\$[\mathbf{A}] + \beta\$[\mathbf{B}]$
- $\$ is completely positive: $\$ \otimes I_B$ is positive for all extensions B.

Purification Simplification



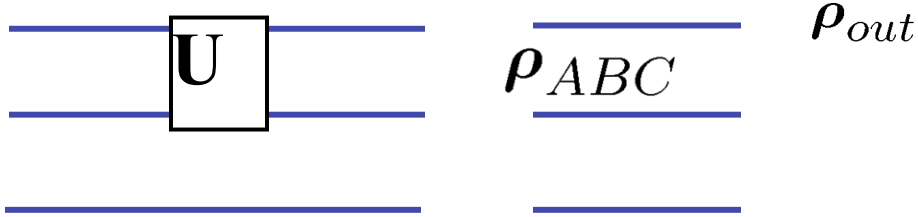
$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$



Purification of system B

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C$

$|\nu\rangle = |\psi_{BC}\rangle$

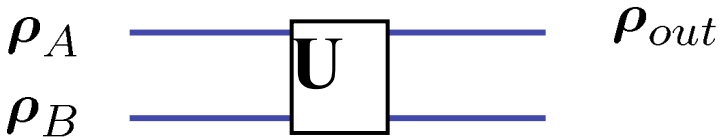


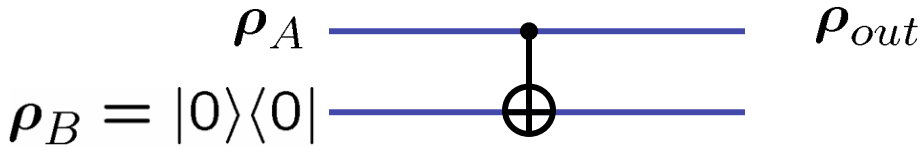
$\rho_{out} = \sum_k \mathbf{A}_k \rho_A \mathbf{A}_k^\dagger$ where $\mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle$

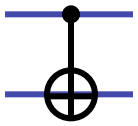
Purification of system B

$\rho_{out} = \sum_k \mathbf{A}_k \rho_A \mathbf{A}_k^\dagger$ where $\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle$

OSR Example







$$= |0\rangle\langle 0| \otimes \mathbf{I} + |1\rangle\langle 1| \otimes \mathbf{X} = \frac{1}{2}(\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX})$$

$$\boldsymbol{\rho}_B = \textstyle\sum_\nu p_\nu |\nu\rangle\langle\nu| \rightarrow \boldsymbol{\rho}_B = |0\rangle\langle 0|$$

$$\mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle \rightarrow \mathbf{A}_\mu = \langle\mu|\mathbf{U}|0\rangle$$

$$\mathbf{A}_0 = \langle 0|\mathbf{U}|0\rangle = \langle 0|\frac{1}{2}[\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX}]|0\rangle = \frac{1}{2}(\mathbf{I} + \mathbf{Z}) = |0\rangle\langle 0|$$

$$\mathbf{A}_1 = \langle 1|\mathbf{U}|0\rangle = \langle 1|\frac{1}{2}[\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX}]|0\rangle = \frac{1}{2}(\mathbf{I} - \mathbf{Z}) = |1\rangle\langle 1|$$

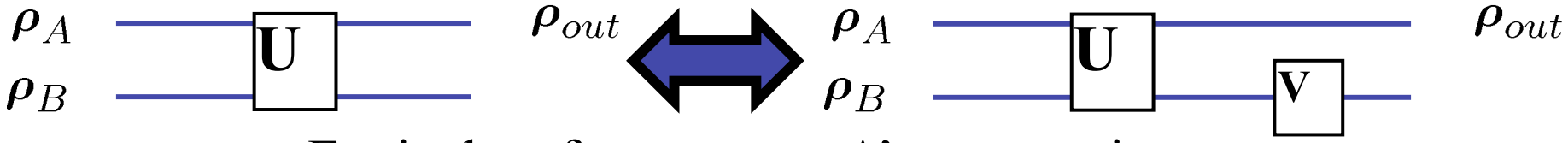
$$\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_A \mathbf{A}_k^\dagger \rightarrow \boldsymbol{\rho}_{out} = |0\rangle\langle 0|\boldsymbol{\rho}_A|0\rangle\langle 0| + |1\rangle\langle 1|\boldsymbol{\rho}_A|1\rangle\langle 1|$$

OSR Unitary Freedom

OSR: $\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle$ $\quad$ $\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_{in} \mathbf{A}_k^\dagger$

Whose $|k\rangle$?

$|k\rangle$ was from Trace operation: basis arbitrary.



Equivalent from system A's perspective

$$|l\rangle = \sum_k V_{kl}|k\rangle, |k\rangle = \sum_l V_{kl}^*|l\rangle, \quad VV^\dagger = I$$

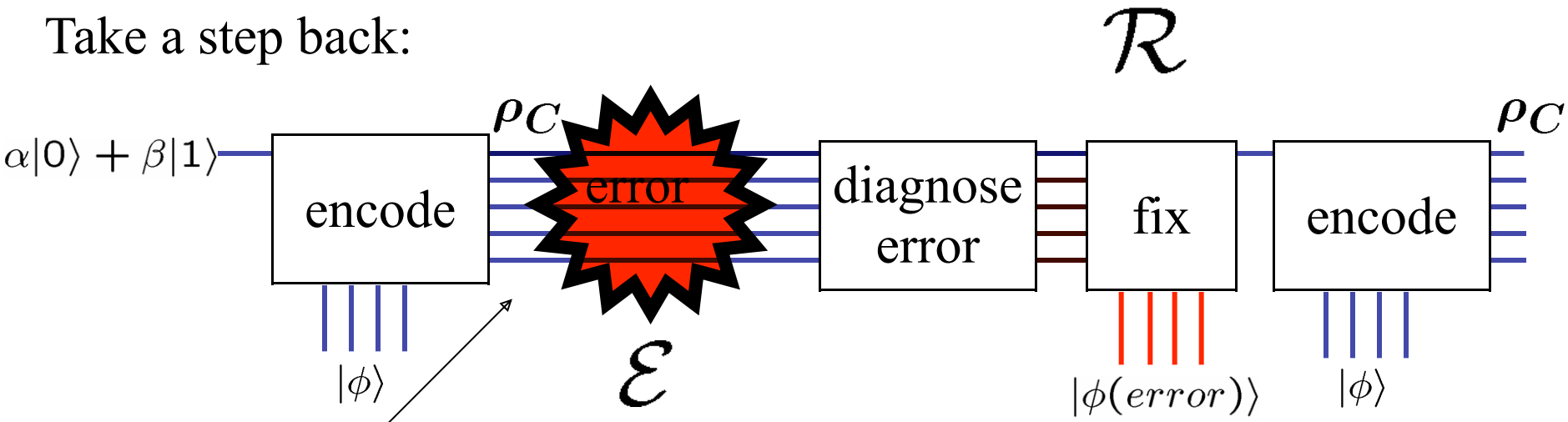
New OSR: $\mathbf{B}_l = \langle l|\mathbf{U}|\nu\rangle$

$$\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle = \sum_l V_{kl}\langle l|\mathbf{U}|\nu\rangle = \sum_l V_{kl}\mathbf{B}_l$$

Two OSR's $\{\mathbf{A}_k\}$ and $\{\mathbf{B}_k\}$ are equivalent iff there exists a unitary matrix V_{kl} such that $\mathbf{A}_k = \sum_l V_{kl}\mathbf{B}_l$.

Do the Hokey Pokey

Take a step back:



Here quantum information has simply been encoded into ρ_C

Then an error occurs

$$\mathcal{E}[\rho_C] = \sum_k \mathbf{E}_k \rho_C \mathbf{E}_k^\dagger$$

Then a recovery procedure

$$\mathcal{R}[\mathcal{E}[\rho_C]] = \sum_l \mathbf{R}_l \sum_k \mathbf{E}_k \rho_C \mathbf{E}_k^\dagger \mathbf{R}_l^\dagger$$

An miraculously we recover the encoded state ρ_C

$$\mathcal{R}[\mathcal{E}[\rho_C]] = \rho_C$$

Lets Get Definitional

A quantum code C is simply a ~~subspace~~ subsystem

Quantum Error Correcting Code

A quantum error correcting code C for the quantum operation $\mathcal{E}$ is a code for which there exists a recovery operator $\mathcal{R}$ such that $\mathcal{R} \circ \mathcal{E}$ acts proportional to identity on the codespace.

Iff Condition for Quantum Error Correcting Code

A code C with orthonormal basis $\{|i\rangle\}$ is a quantum error correcting code for the quantum operation $\mathcal{E}$ with operation elements $\{\mathbf{E}_k\}$ iff

$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

In Action

$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

$|0_L\rangle = |000\rangle$ error set $\mathbf{E}_a \in \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3\}$

$|1_L\rangle = |111\rangle$ $\mathbf{E}_b^\dagger \mathbf{E}_a \in \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_1\mathbf{X}_2, \mathbf{X}_1\mathbf{X}_3, \mathbf{X}_2\mathbf{X}_3\}$

If $\mathbf{E}_b^\dagger \mathbf{E}_a \neq \mathbf{I}$, $\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = 0$

$|0_L\rangle = |000\rangle$ error set $\mathbf{E}_a \in \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{Z}_1\}$

$|1_L\rangle = |111\rangle$

$\mathbf{E}_b^\dagger = \mathbf{I}$, $\mathbf{E}_a = \mathbf{Z}_1$, $\langle 0_L|\mathbf{Z}_1|0_L\rangle = 1$ but $\langle 1_L|\mathbf{Z}_1|1_L\rangle = -1$

Diagonal Error Condition

Error correcting code criteria: $\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = C_{ab}\delta_{ij}$

$$C_{ab}\delta_{ij} = \langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = \left(\langle j|\mathbf{E}_b^\dagger \mathbf{E}_a|i\rangle\right)^* = C_{ba}^*\delta_{ij}$$

$C_{ab} = C_{ba}^*$: C is Hermitian

Change of OSR basis: $\mathbf{F}_k = \sum_k u_{ka}\mathbf{E}_a$

Error correcting code criteria: $\langle i|\mathbf{F}_k^\dagger \mathbf{F}_l|j\rangle = D_{kl}\delta_{ij}$

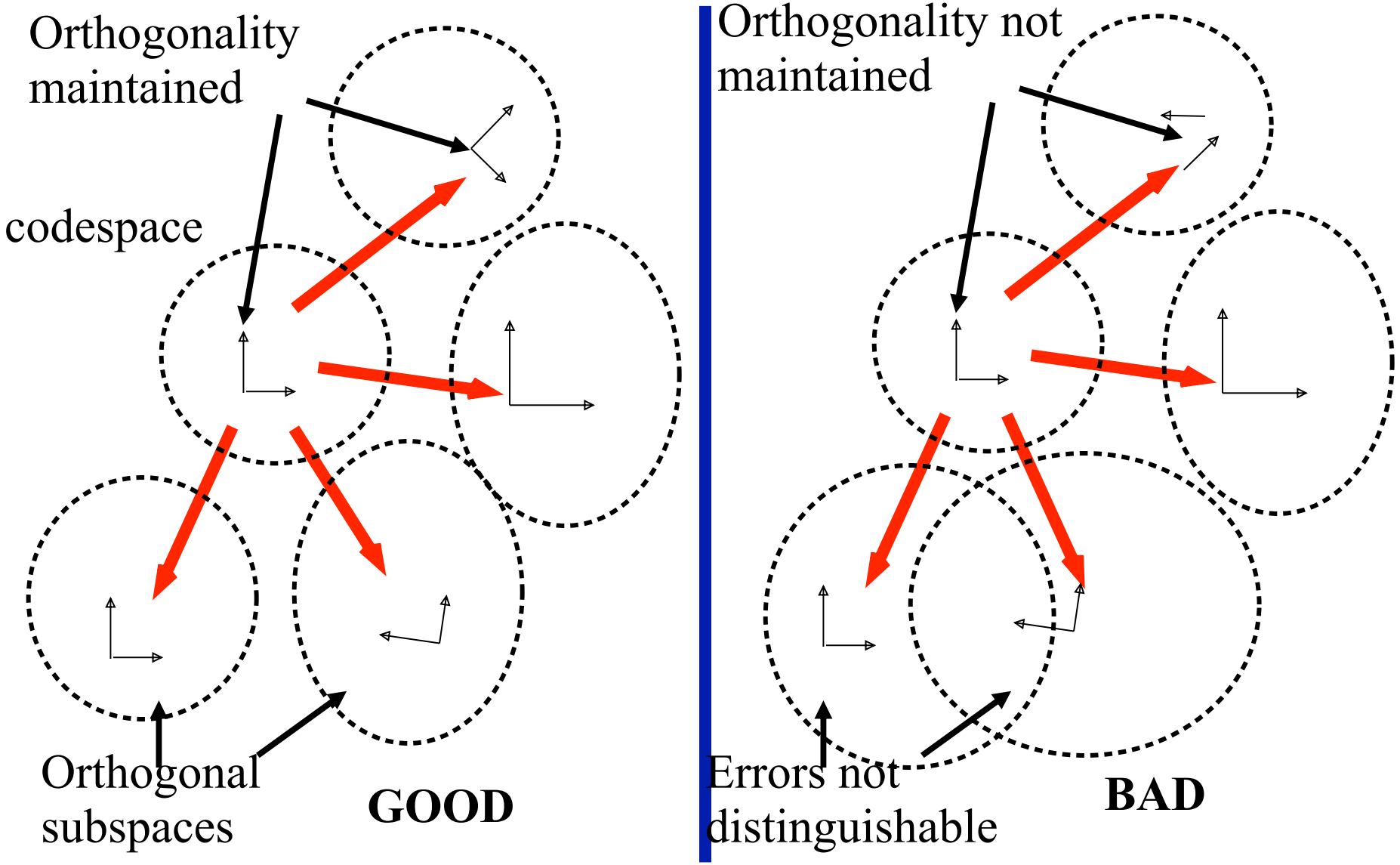
$$\langle i|\mathbf{F}_k^\dagger \mathbf{F}_l|j\rangle = \sum_{ab} u_{ka}^* u_{lb}\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = \sum_{ab} u_{ka}^* C_{ab} u_{bl}\delta_{ij}$$

$$\sum_{ab} u_{ka}^* C_{ab} u_{bl} = D_{kl} = d_k\delta_{kl}$$

$$\boxed{\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = d_a\delta_{ab}\delta_{ij}}$$

Pictures

$$\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = D_a \delta_{ab}\delta_{ij}$$



1994 Reasons to be a Pessimist

Continuous set of errors:

Classical errors on digital information are discrete.

Quantum errors on classical information are continuous.

So far we have designed our error correction for a specific OSR evolution

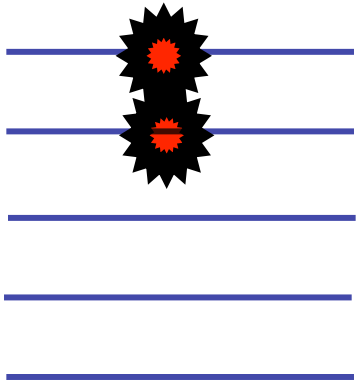
$$\boldsymbol{\rho} \to \sum_a \mathbf{F}_a \boldsymbol{\rho} \mathbf{F}_a^\dagger$$

such that for our recover operator $\mathcal{R}$

$$\mathcal{R}\left[\sum_a \mathbf{F}_a \boldsymbol{\rho}_C \mathbf{F}_a^\dagger\right] \propto \boldsymbol{\rho}_C$$

code corrects OSR operators $\{\mathbf{E}_a\} \to$ code corrects OSR operators $\{\sum_b v_{ba}\mathbf{E}_a\}$.

How Have We Helped?



Suppose error acts on one of many qubits

$$\boldsymbol{\rho} \to \sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger, \quad \mathbf{E}_k = \mathbf{E}_k \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Single qubit error correcting code:

$$\text{If } \mathbf{E}_k = e_0 \mathbf{I} + \textstyle\sum_\alpha e_\alpha \boldsymbol{\sigma}_\alpha, \text{ then } \mathcal{R}\left[\sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger\right] = \boldsymbol{\rho}$$

$$\mathcal{R}[\boldsymbol{\rho}] = \sum_a \mathbf{U}_a^\dagger \mathbf{P}_a \boldsymbol{\rho} \mathbf{P}_a \mathbf{U}_a$$

Notion of "no error outcome" for a single qubit error, i.e. the measurement projector $\mathbf{P}_0 = \mathbf{P}_C$, implies "probability of no error" $= 1 - p$.

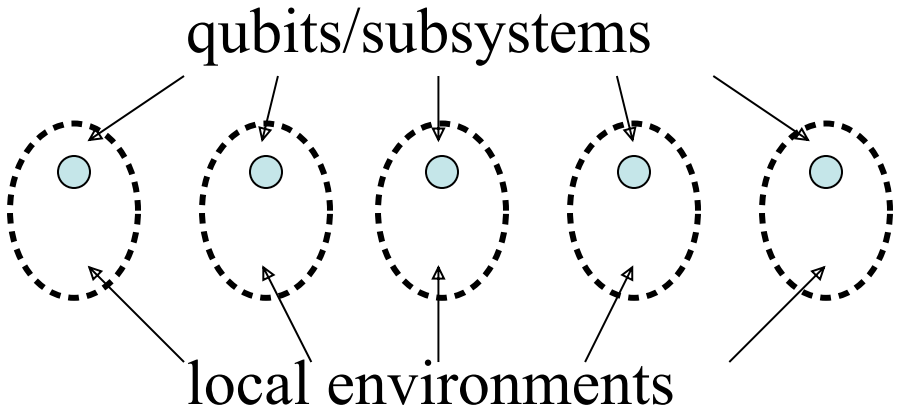
Now suppose two errors act "independently":

$$\boldsymbol{\rho} \to \sum_l \mathbf{F}_l \sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger \mathbf{F}_l^\dagger, \quad \mathbf{E}_k = \mathbf{E}_k \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}, \mathbf{F}_l = \mathbf{I} \otimes \mathbf{F}_l \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Notion of no two error probability $= (1-p)^2$

Independent Error Model

WARNING: "An approximation only a physicist could love."



Evolution dominated by interaction with different environment modes whic
do not interact with each other:

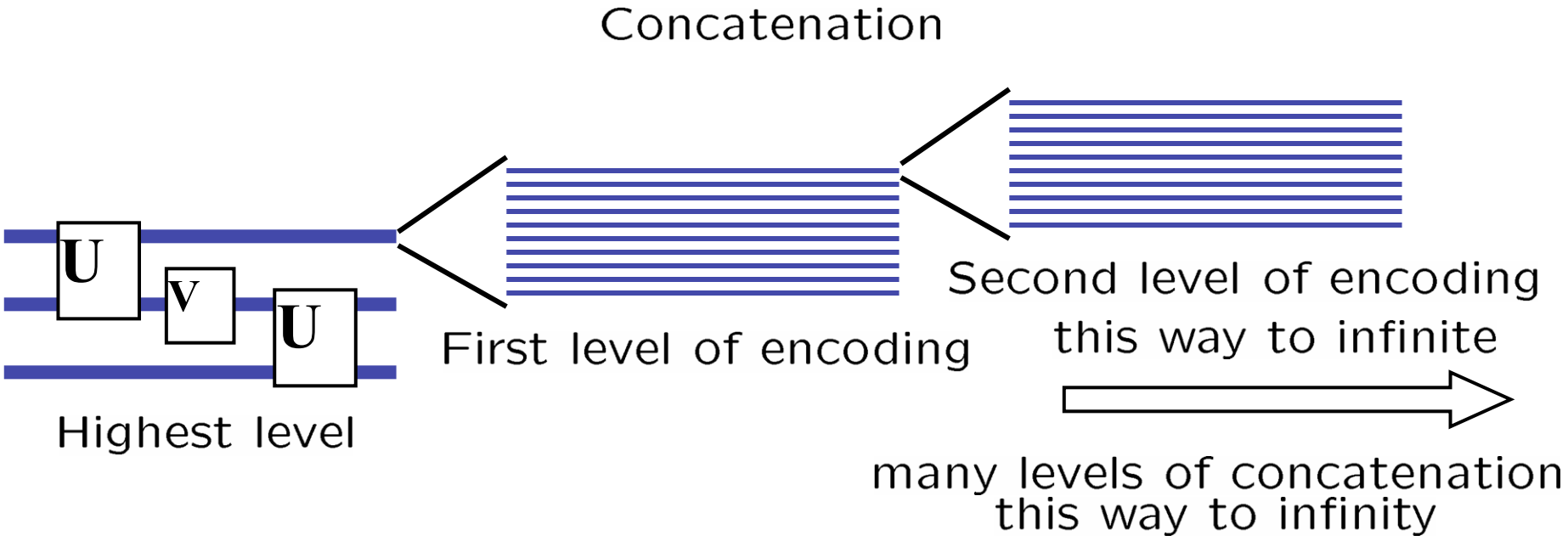
$$\mathcal{E}[\boldsymbol{\rho}] = \sum_{k_1,k_2,\cdots,k_n} \mathbf{E}_{k_1} \otimes \mathbf{E}_{k_2} \otimes \cdots \otimes \mathbf{E}_{k_n} \boldsymbol{\rho} \mathbf{E}_{k_1}^{\dagger} \otimes \mathbf{E}_{k_2}^{\dagger} \otimes \cdots \otimes \mathbf{E}_{k_n}^{\dagger}$$

Notion of "no error probability" gives credence to standard argument for
error correction success.

Really we can weaken this requirement (exponentially decreasing correlati
[time and space] between errors).

Perfection Through Concatenation

Idea 2: Error correct these encoded qubits to fix 'errors'



Threshold Theorem for Quantum Memory

If the single qubit error rate is below some threshold p_t, then we can store quantum states to accuracy ϵ using poly$(\log(\epsilon^{-1}))$ qubits.

Review



Rolf Landauer
IBM

- Maintenance of giganto-coherence?



By encoding quantum information we can preserve coherence

Iff Condition for Quantum Error Correcting Code

$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

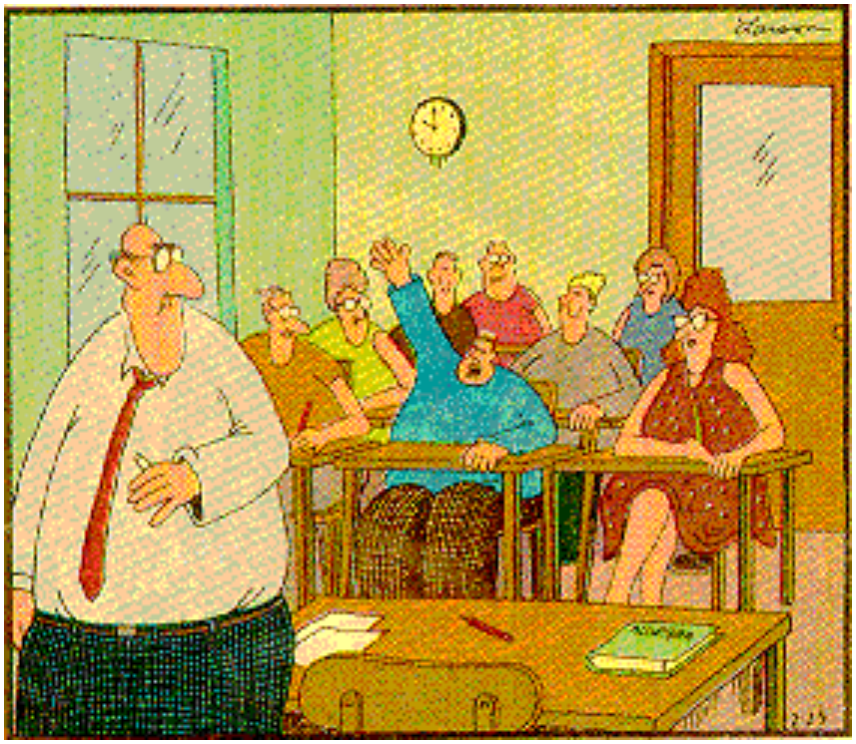
"fight entanglement with entanglement"

The Next Episode

Actual error correcting codes!

Stabilizer formalism and codes

Topological quantum computation



"Dave, may I be excused? My brain is full."

